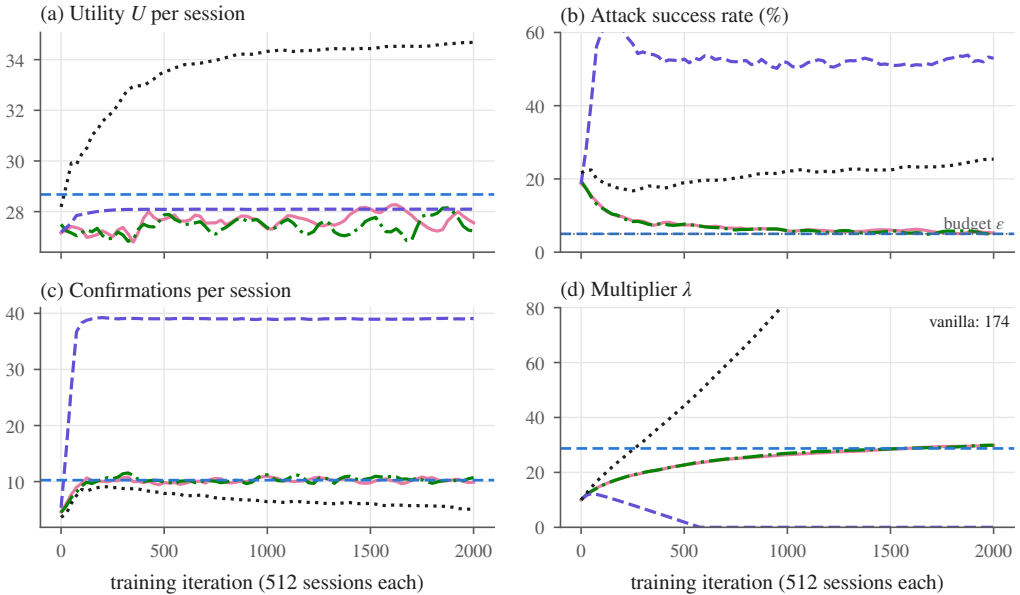
(a) Utility U per session
34
32
30
28
(b) Attack success rate (%)
60
40
20
0
budget ε
(c) Confirmations per session
40
30
20
10
0
500
1000
1500
2000
training iteration (512 sessions each)
(d) Multiplier λ
80
60
40
20
0
vanilla: 174
0
500
1000
1500
2000
training iteration (512 sessions each)
Alg. 1, fixed profile
Alg. 1, randomized profiles
Alg. 1, fatigue-free simulator
Alg. 1 with per-group std (vanilla GRPO)
CMDP optimum (DP, $\lambda^\star$)